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StreamCatTools: An R package for working with StreamCat and LakeCat watershed data in R

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Summary

`StreamCatTools` provides functions for easily working with, visualizing and analyzing StreamCat (Hill, Weber, Leibowitz, Olsen, & Thornbrugh, 2016) and LakeCat (Hill, Weber, Debbout, Leibowitz, & Olsen, 2018) watershed metrics within **R**. The StreamCat and LakeCat datasets provide hundreds of landscape metrics for both the local catchment (e.g. landscape draining to a particular stream reach) and full watershed for every stream reach and lake depicted in the medium resolution National Hydrography Dataset Plus Version 2.1 (NHDPlusV21)(McKay et al., 2012) for the contiguous United States (CONUS). `StreamCatTools` functions wrap the application programming interface (API) for the StreamCat and LakeCat online database and facilitate simple, straightforward access and use of these datasets within R.

Statement of Need

Easily accessible, robust, and consistent watershed data is an underpinning of hydrology research, water quality monitoring programs, and predictive modelling applications, to

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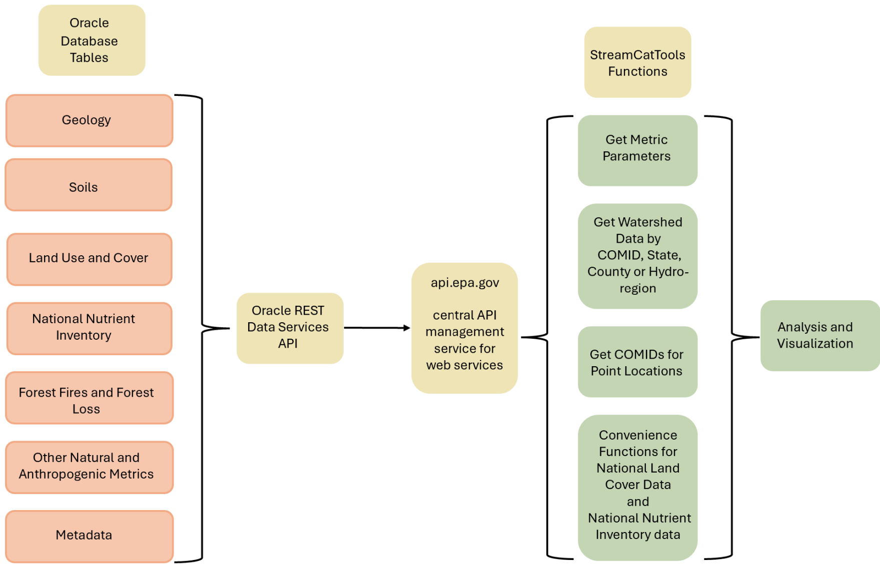


Figure 1: Diagram of the StreamCat and StreamCatTools framework. A backend Oracle database and web service are exposed through api.gov to functions in StreamCatTools which simplify accesss and analysis of the data in R via the application programming interface.

name just a few examples. The StreamCat (Hill et al., 2016) and LakeCat (Hill et al., 2018) datasets fill this need by providing nationally consistent curated watershed data for CONUS that has had stringent quality control applied. The data encompasses hundreds of watershed metrics for every stream reach and lake feature represented in the National Hydrography Dataset Plus Version 2.1 (NHDPlusV21) (McKay et al., 2012). StreamCatTools fills the need for easily accessible watershed metrics for CONUS by: (1) providing a simple interface in the R programming language to the StreamCat and LakeCat web services, (2) providing convenient functionality to find available StreamCat and LakeCat metric names and information about variables, and (3) extracting StreamCat and LakeCat metrics by COMID (a unique identifier in the NHDPlusV2 framework), by state, by county, by NHD Hydro-region, or for all of CONUS. Providing this valuable watershed data via web services in R follows the FAIR principles laid out in (Wilkinson et al., 2016).

Package Overview

`StreamCatTools` provides a simple streamlined set of functions to easily query and ingest watershed landscape metrics into an R session. Figure 1 shows the overall framework of the StreamCat database, application programming interface, and functionality in the package that simplifies data access in R using web services for the StreamCat and LakeCat datasets.

The core functions in `StreamCatTools` leverage the `httr2` library (Wickham, 2025) for a modern, pipeable method for working with APIs. Specifically, `StreamCatTools` simplifies the calls to the API for StreamCat and LakeCat data within the R programming language to allow a user to: (1) access details on available StreamCat and LakeCat metrics, (2) match users' sites to NHDPlusV21 waterbodies (streams or lakes) to access StreamCat and Lakecat data, and (3) retrieve StreamCat and LakeCat watershed metrics by COMID, by state, by county, by hydroregion, or for all of CONUS. Additionally, several convenience functions retrieve data from particular years for data with time series, such as the National

Land Cover Database (NLCD) (USGS, 2024) and the National Nutrient Inventory (NNI) (Brehob et al., 2025). Lastly, there is plotting functionality for use with NNI metrics as well as a function to retrieve the watersheds for specific lake features as an `sf` object. Users can likewise access watersheds for specific COMIDs using the `nhdplusTools` package for visualization of and plotting of watersheds to go along with StreamCat or LakeCat metrics.

You can install the most recent version of `StreamCatTools` from CRAN by running:

```
# install.packages("StreamCatTools")
```

You can install the most recent development version from GitHub by running:

```
# library(remotes)
# install_github("USEPA/StreamCatTools", build_vignettes=FALSE)
```

`StreamCatTools` is loaded into an **R** session:

```
library(StreamCatTools)
```

`StreamCatTools` includes several functions to facilitate accessing and working with the data. First, users can list metric names and find out more about StreamCat and LakeCat data available in StreamCat using using the `sc_get_params` and `lc_get_params` functions. The parameters `aoi` and `metric_names` provide a user with the available areas of interest for metrics and names of metrics. The areas of interest available for all metrics in StreamCat and Lakecat are 'cat' and 'ws' (catchment and watershed) and in StreamCat some metrics are available at the 100 meter riparian buffer scale ('catrp100' and 'wsrp100') and for certain metrics the designation of 'other' needs to be used (where the metric is in-stream or derived and not part of the other listed areas of interest). Extracting this information in `StreamCatTools` looks like this:

```
aois <- sc_get_params(param='aoi')

name_params <- sc_get_params(param='metric_names')

print('areas of interest include: ')
```

```
## [1] "areas of interest include: "
```

```
cat(paste0(aois,collapse = "\n"))
```

```
## cat
## catrp100
## other
## ws
## wsrp100
```

```
print('A selection of available StreamCat metrics include: ')
```

```
## [1] "A selection of available StreamCat metrics include: "
```

```r
cat(paste0(name_params[1:10],collapse = "\n"))
```

```
## agkffact
## al2o3
## bankfulldepth
## bankfullwidth
## bfi
## canaldens
## cao
## cbnf
## chem
## clay
```

And the same information for LakeCat metrics is derived in similar fashion:

```r
aois <- lc_get_params(param='aoi')

name_params <- lc_get_params(param='metric_names')

print('areas of interest include: ')
```

```
## [1] "areas of interest include: "
```

```r
cat(paste0(aois, collapse = "\n"))
```

```
## cat
## ws
```

```r
print('A selection of available LakeCat metrics include: ')
```

```
## [1] "A selection of available LakeCat metrics include: "
```

```r
cat(paste0(name_params[1:10],collapse = "\n"))
```

```
## agkffact
## al2o3
## bfi
## canaldens
## cao
## cbnf
## clay
## coalminedens
## compstrgth
## damdens
```

StreamCat and LakeCat are built around the concepts of local drainage area (i.e catchment) and watershed (i.e. the local drainage area and all upstream catchments) (Hill et al., 2016). This approach uses the NHDPlusV21 hydrographic framework of catchments as the building block for summarizing landscape information represented in landscape data and then aggregating both the catchment landscape summary and all upstream catchments using the weighted average of metrics in most cases, but using a sum or count for certain metrics. `sc_get_params` and `lc_get_params` also include a 'variable_info' parameter to return more detailed metadata for metrics including both short and long metric descriptions, years available (if applicable), units, and the metric category.

```
var_info <- sc_get_params(param='variable_info')

my_data <- head(var_info[c(5,10,29,33,45,102),c('metric','short_description',
↪  'year','units')])

knitr::kable(
  my_data,
  format = "latex",
  booktabs = TRUE,
  longtable = TRUE,
  col.names = c('Metric', 'Short Description', 'Year', 'Units'),
  align = "llll"
)  |>
  kableExtra::kable_styling(
    latex_options = c("repeat_header"),
    font_size = 7
  )  |>
  kableExtra::column_spec(1, width = "1.0in")  |>
  kableExtra::column_spec(2, width = "1.0in")  |>
  kableExtra::column_spec(3, width = "1.0in")  |>
  kableExtra::column_spec(4, width = "1.0in")
```

Metric	Short Description	Year	Units
agkffact[AOI]	Ag Soil Erodibility Kf Factor	NA	Unitless
bfi[AOI]	Base Flow Index	NA	Percent
huden[Year][AOI]	Mean Housing Density	2010	Count/Square Kilometer
inorgnwetdep[Year][AOI]	Mean Annual Precipitation-Weighted Nitrogen	2008	Kilogram/Hectare/Year
n_ags_[Year][AOI]	Nitrogen Agricultural Surplus	1987-2017	Kilograms
pcthighsev[Year][AOI]	Percent High Burn Severity Class For Year	1984-2018	Percent

Additional functions provided for getting metadata about the underlying StreamCat and LakeCat data include the `sc_fullname` and `lc_fullname` functions to provide the descriptive full name for any given metric in StreamCat or LakeCat:

```
fullname <- sc_fullname(metric='pctgrs2019')
fullname
```

```
## [1] "Grassland/Herbaceous Percentage 2019"
```

Users can also filter metric names and information by the metric year(s), the indicator categories for metrics, the metric dataset names, or the areas of interest available for a given metric using the `sc_get_metric_names` or `lc_get_metric_names` functions:

```r
metrics <- sc_get_metric_names(category = c('Deposition','Climate'),
                               aoi=c('Cat','Ws'))
my_data <- head(metrics[,c('Category','Metric','AOI')],10)
knitr::kable(
  my_data,
  format = "latex",
  booktabs = TRUE,
  longtable = TRUE,
  col.names = c('Category','Metric','AOI'),
  align = "lll"
)  |>
  kableExtra::kable_styling(
    latex_options = c("repeat_header"),
    font_size = 7
  )  |>
  kableExtra::column_spec(1, width = "1.2in") |>
  kableExtra::column_spec(2, width = "1.4in") |>
  kableExtra::column_spec(3, width = "1.0in")
```

Category	Metric	AOI
Climate	bfi[AOI]	Cat, Ws
Deposition	inorgnwetdep[Year][AOI]	Cat, Ws
Deposition	nh4[Year][AOI]	Cat, Ws
Deposition	no3[Year][AOI]	Cat, Ws
Climate	precip8110[AOI]	Cat, Ws
Climate	precip9120[AOI]	Cat, Ws
Climate	precip[Year][AOI]	Cat, Ws
Deposition	sn[Year][AOI]	Cat, Ws
Climate	tmax8110[AOI]	Cat, Ws
Climate	tmax9120[AOI]	Cat, Ws

More details on these functions can be found at the package introduction page.

The primary package functionality is in the `sc_get_data` and `lc_get_data` functions which allow users to extract catchment or watershed metrics of interest by providing NHDPlusV21 COMIDs for streams or lakes within the database. Users can also request data for a given state(s), county(ies), hydroregion(s), or all of CONUS. Additionally, convenience functions are provided for accessing the NLCD and NNI datasets, respectively, using `sc_nlcd` and `lc_nlcd` and `sc_nni` and `lc_nni`.

The following example shows a request for two catchment and watershed metrics for three stream reaches:

```r
df <- sc_get_data(metric='pcturbmd2019,damdens',
                  aoi='cat,ws',
                  comid='179,1337,1337420')
```

The first line requests percent of NLCD medium intensity developed land cover in 2019 and the density of dams. The second line specifies that the geographic area of interest (aoi) is the drainage to the local reach scale (i.e. 'cat', short for catchment) and the full watershed. Finally, the final line request this data for three NHDPlusV2 stream segments specified by their unique COMIDs.

Applications and Discussion

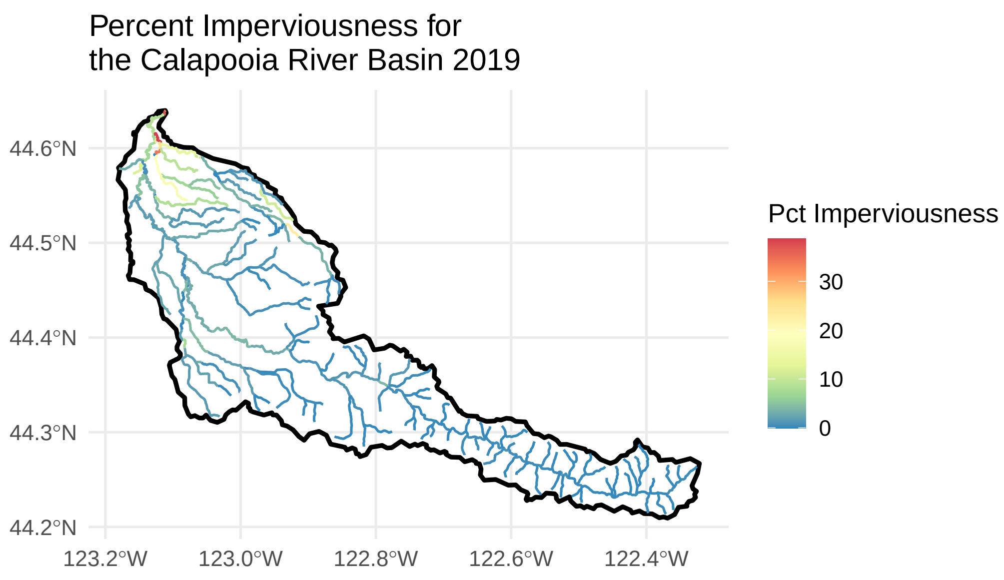
Watershed metrics from `StreamCatTools` can be easily visualized with functions from `nhdplusTools` (Blodgett & Johnson, 2023) and `ggplot2` (Wickham, 2016). The example below plots the NLCD percent imperviousness for the the local drainage (catchment in NHDPlusV2 syntax) and displays the values mapped to each stream reach and to the overall basin boundary:

```
library(nhdplusTools)
library(ggplot2)
library(ggspatial)
library(StreamCatTools)

start_comid = 23763517
nldi_feature <- list(featureSource = "comid", featureID = start_comid)
flowline_nldi <- nhdplusTools::navigate_nldi(nldi_feature, mode = "UT",
↪  data_source = "flowlines", distance=5000)
df <- sc_get_data(metric='pctimp2019', aoi='cat',
↪  comid=flowline_nldi$UT_flowlines$nhdplus_comid)
flowline_nldi <- flowline_nldi$UT_flowlines
flowline_nldi$PCTIMP2019 <- df$pctimp2019cat[match(flowline_nldi$nhdplus_comid,
↪  df$comid)]
basin <- nhdplusTools::get_nldi_basin(nldi_feature = nldi_feature)

calapooia <- ggplot() +
    geom_sf(data = basin,
            fill = NA,
            color = "black",
            linewidth = 1) +
    geom_sf(data = flowline_nldi,
            aes(colour = PCTIMP2019)) +
    scale_y_continuous() +
    scale_color_distiller(palette = "Spectral") +
    labs(color = "Pct Imperviousness") +
    theme_minimal(12) +
    ggtitle('Percent Imperviousness for \nthe Calapooia River Basin 2019')

plot(calapooia)
```



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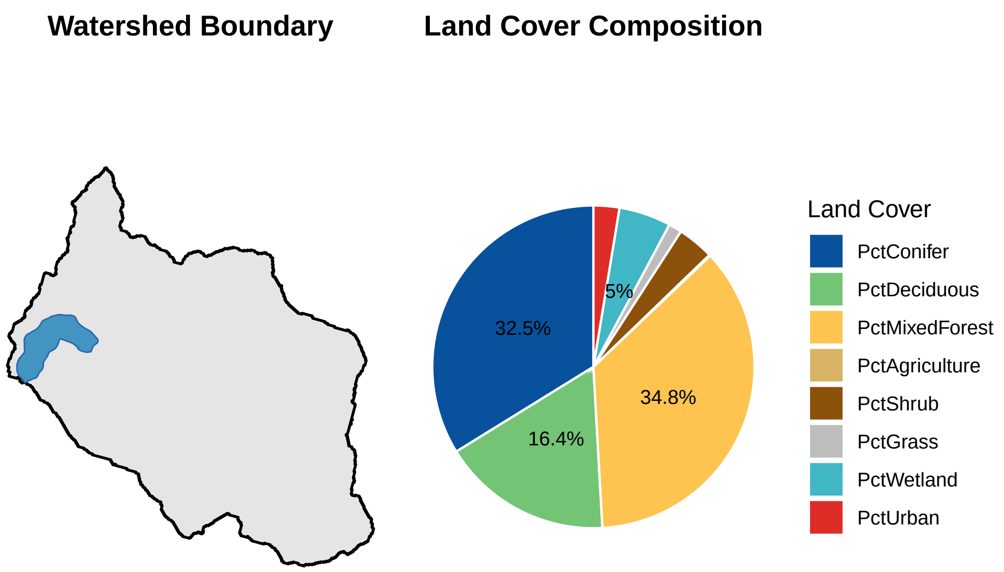
Watersheds for lakes can also be retrieved using the `lc_get_watershed` function in order to visualize LakeCat metrics along with the plotted watersheds for lake features:

```
library(ggplot2)
library(patchwork)
library(ggforce)

df <- lc_get_nlcd(comid='19334077', year='2019', aoi='ws')
lake <- nhdplusTools::get_waterbodies(id = 19334077)
ws <- lc_get_watershed(comid = 19334077, huc2 = "01",huc2_filter = "01",
                      threads = 2,retries = 5)
```

```
##   |
##   |
##   |
##   |
## [10:28:40] Rows returned: 1
##   |
```

```
df <- df |>
  dplyr::mutate(
    PctConifer = pctconif2019ws,
    PctDeciduous = pctdecid2019ws,
    PctMixedForest = pctmxfst2019ws,
    PctAgriculture = pctcrop2019ws + pcthay2019ws,
    PctShrub = pctshrb2019ws,
    PctGrass = pctgrs2019ws,
    PctWetland = pcthbwet2019ws + pctwdwet2019ws,
    PctUrban = pcturbhi2019ws + pcturblo2019ws + pcturbmd2019ws + pcturbop2019ws
  ) |>
  dplyr::select(PctConifer, PctDeciduous, PctMixedForest, PctAgriculture,
↪   PctShrub, PctGrass, PctWetland, PctUrban)

landcover_df <- df |>
  tidyr::pivot_longer(
    cols = c(PctConifer, PctDeciduous, PctMixedForest, PctAgriculture,
             PctShrub, PctGrass, PctWetland, PctUrban),
    names_to = "category",
    values_to = "value"
  ) |>
  dplyr::mutate(
    category = factor(category,
      levels = c("PctConifer","PctDeciduous","PctMixedForest","PctAgriculture",
                 "PctShrub","PctGrass","PctWetland","PctUrban")
    ),
    pct_label = ifelse(value >= 5, paste0(round(value, 1), "%"), "")  # blank
    ↪   label if < 5%
  )

pie_colors <- c(
  PctConifer      = "#08519c",
  PctDeciduous    = "#74c476",
  PctMixedForest  = "#fec44f",
  PctAgriculture  = "#d8b365",
  PctShrub        = "#8c510a",
  PctGrass        = "#bdbdbd",
  PctWetland      = "#41b6c4",
  PctUrban        = "#de2d26"
```

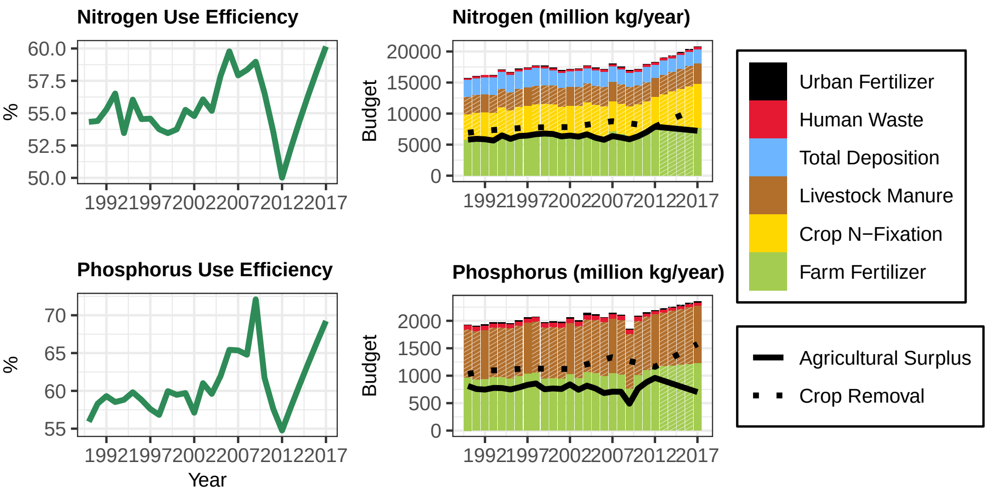
```
)

# --- map ---
p_map <- ggplot() +
  geom_sf(data = ws, fill = "grey90", color = "black", linewidth = 0.6) +
  geom_sf(data = lake, fill = "#4393c3", color = "#2166ac", linewidth = 0.3) +
  theme_void() +
  labs(title = "Watershed Boundary") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

# --- pie ---
p_pie <- ggplot(landcover_df, aes(x = "", y = value, fill = category)) +
  geom_col(width = 1, color = "white") +
  coord_polar(theta = "y") +
  geom_text(aes(label = pct_label), position = position_stack(vjust = 0.5), size
  ↪ = 3, color = "black") +
  scale_fill_manual(values = pie_colors, name = "Land Cover") +
  theme_void() +
  labs(title = "Land Cover Composition") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

p_map + p_pie + plot_layout(ncol = 2, widths = c(1, 1))
```

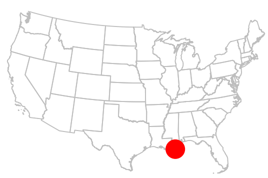


Plotting functions to specifically plot NNI metrics in StreamCat have been developed and are also available (Markley et al., 2026) such as the following example to plot annual time series of nitrogen and phosphorus budget data for a given watershed such as the Mississippi-Atchafalaya River Basin:

```
library(StreamCatTools)
library(ggplot2)
library(ggpattern)
com <- '22812041'
sc_plotnni(comid = com, include.nue = TRUE)
```

```
## If the plot does not render to the plot window when calling the function either

## Retrieving data for the year 2024
```





Future functionality for `StreamCatTools` includes expanding the scope of plotting functions as well as expanding the range of metrics and ease of querying these metrics. `StreamCatTools` currently facilitates easy ingestion of StreamCat and LakeCat watershed landscape metrics into workflows in R which is of great use to state watershed planners, researchers and non-governmental organizaitons and borne out by the over 5000 package downloads and over 350 citations of the underlying StreamCat and LakeCat data served by the `StreamCatTools` package.

Acknowledgements

Examples of using StreamCat and LakeCat make extensive use of `nhdplusTools`(Blodgett & Johnson, 2023) and the functions for accessing the API are facilitated through use of `httr2`(Wickham, 2025). Figures were created using `ggplot2` (Wickham, 2016).

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